1.

Train 4 models, and evaluate on unrotated MNIST / rotated MNIST

- CNN → unrotated MNIST / rotated
- C_4 ENN → urotated / rotated

C_4 ENN : C_4: C_4 group   ENN : equariance neural network

MNIST : a hand written number of pictures of data equipped with the label labeling the number of the picture. Both train and test data included.

rotated : before train / test, the picture was rotated in advance.

evalate:

Evaluate trained model, test the accuracy of correctly distinguish number given test picture.

footnote: reason for C_4: for the grid $\mathbb{Z}_2$ only subgroup of D_4 are exact symmetries, ref … chav. 2.8.

(note: these parts are necessary background, in the actual report can be omitted some parts accordingly.)

Result: (picture)

△ Main result:

① Compare CNN (train rot) and C_4 ENN (train rot) on rotated MNIST. $\to$ 96.65% / $\to$ 97.32%

We find C_4 steerable has the better accuracy which means the inherted rotating symmetry in network does makes models perform beter in distinguish rotated data.

△ other comparisions (these will be too long for 5 mins report, only listed as the completeness of disscussion, in practice one can skip or quickly go through this part.)

② unrot CNN $\xrightarrow{\text{test}}$ rotated, 46.59 %

unrot C4 $\xrightarrow{\text{test}}$ rotated, 84.99 %

CNN performs awful, which means it barely learn nothing on the rotated feature of number,

C4 ENN still performs pretty good, which indicates it doesn't learn from rotated feature, but inhert the rotated feature in the construction (for more info. refer to ... char. 2)

⑤ unrotated CNN $\xrightarrow{\text{test, unrotated}}$ 99.45%

unrot C4 ENN 97.91%

the compulsory applied group restriction on C4 ENN. make less learning parameter, making it in contrast less expresivess on standard task.

⑥ rot CNN $\xrightarrow{\text{test unrotated}}$ 99.66%

rot C4 ENN 96.68%

Benefiting rooted symmetry structure, ENN can put more source to learn the digit it self rather than then rotated feature. While CNN have to seperate some source to learn rotated feature by rote. (Capacity Dilution)

2.

6/9. problem :

Intro :

The ENN suffers from 6/9 problem :

(2 pictures showing 6 and rotated 9)

rotated symmetric ENN can only learn in the sense of rotation.

In the sense of rotation, 6 and 9 looks alike.

Therefore, it is easy for ENN mistake 6 with 9.

footnote : it can be generated to other number, for instance, 5 and 7, to other task rather than tells digits and the boarder symmetric groups rather than rotation.

Solution :

group restriction,

In this case, we first following this first 5 layers the same C_4 structure as traditional C_4 ENN. (i.e, layers are invariant under C_4 group operation (i.e. 90°, 180°, 270°, 360° rotation)

then restricts the group $C_4 \to \{e\}$ in the sixth layer, only require it is invariant under trival group $\{e\}$ symmetry (i.e. no symmetric restriction)

Here are 2 diagrams sketchy the main difference

(footnote①: in partice we also need to claim the rep. of group, and there is a full connection classifer in the end Here for simplicity we omit that pares)

footnote②: for detail of group restriction ref ... chap. 2.7)

① traditional C_4 ENN:

input $\overset{\text{conv.}}{—}$ layer① $\overset{\text{conv.}}{—}$ .... — layer⑥ (output)

input $\leftrightarrow$ output : C_4 opreation invariant

② refined $C_4|_6\{e\}$ ENN:

input $\overset{\text{conv.}}{—}$ layer① $\overset{\text{conv.}}{—}$ ...— layer⑤ — layer⑥ (output)

input $\leftrightarrow$ layor⑤ : C_4 operation invar.

layer⑤ $\leftrightarrow$ layer⑥ : $\{e\}$ operation invar.

Intuitive perspective :

Deeper the layer is, the boarder receptive field is. Therefore, require differeent symm. between diff. depth of layer is actually regarded as require diff. symm. in the different scale. In this Case, we sacrifice the large scale symm. while matain the small scale C_4 symm. for further info. refer ...... 2.7)

(note: for timeing stuff the reporter can selectively skip or quickly go through the technical and the understanding pore. Focus on the intro. and the resule.

key point: group restriction is a technic to mitigate 6/9 problem.)

Result:

figure: resule : (- - -)

Clearly, traditional C_4 ENN suffer from 6/9 problem. (footnote: however not so severe since hand written 6 and 9 still have the difference even in the sense of rotation). While $C_4|_6\{e\}$ and CNN don't have the problem.